

Flagmaker API and examples

CS 141

FlagMaker

We will write nested for loops to draw flags and various other shapes on a 2-dimensional grid

How to run today's examples

1. Download the zipfile [flags-inclass.zip](#)

2. Unzip and double-click

`flags-inclass.code-workspace`

3. copy the appropriate `.jar` file from `tmp` to `lib`

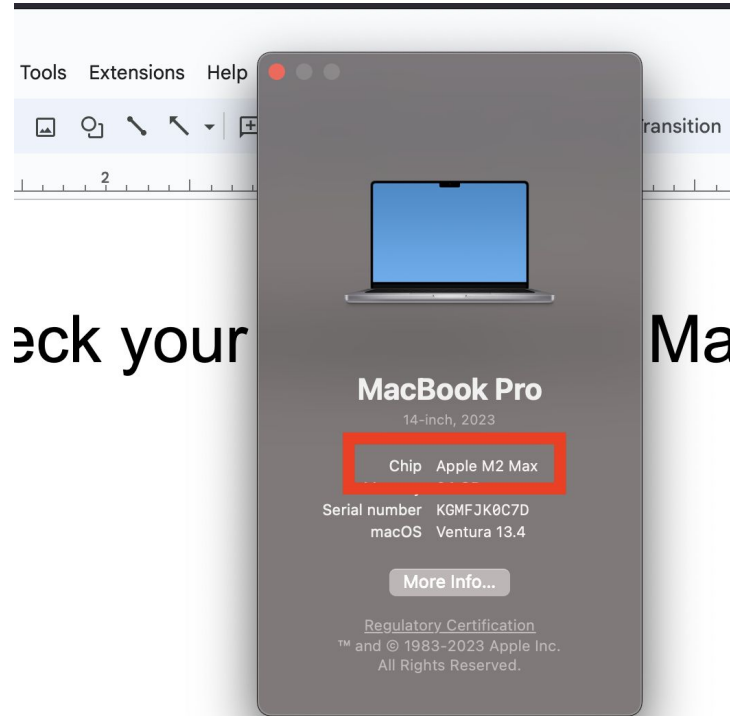
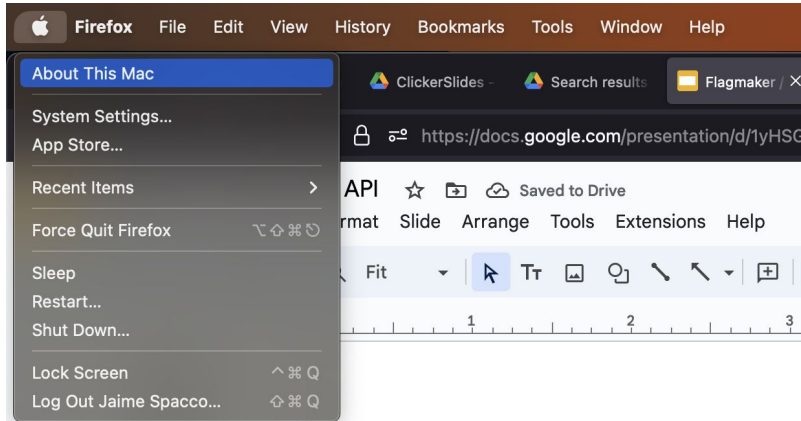
4. run `Main` with the run button

5. read and modify code in `InClass1.java`

Which .jar file is right for you?

- app-windows.jar
- app-linux.jar
- app-mac.jar
 - Mac with Intel CPU
- app-mac_aarch64.jar
 - Mac with Apple Silicon (M1, M2, M3, etc) CPU

To check your hardware on Mac: Apple menu => About This Mac



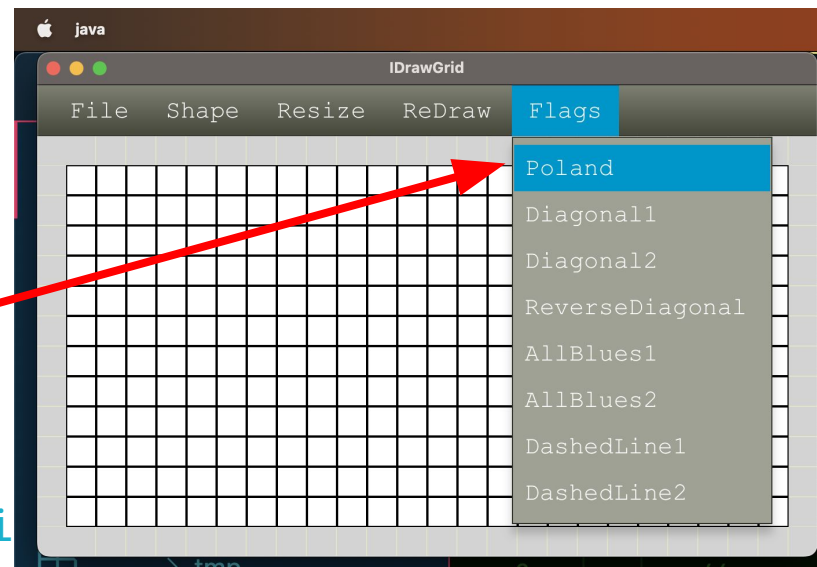
To check your h

Check out InClass1.java

```
@Flag(name = "Poland", num = 1)
public static void drawPoland(IDrawGrid grid)
{
    for (int r=grid.getNumRows()/2; r<grid.getNumRows(); r++)
    {
        for (int c=0; c<grid.getNumCols(); c++)
        {
            grid.setColor(r, c, Color.RED);
        }
    }
}
```

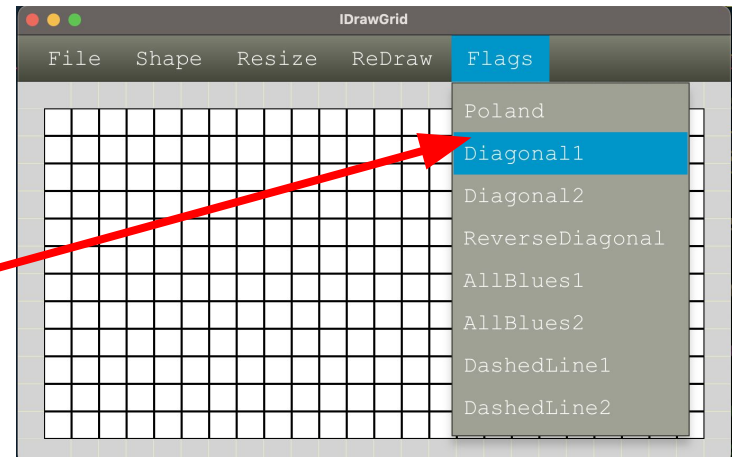
Poland, first in the list

```
@Flag(name = "Poland", num = 1)
public static void drawPoland(IDrawGrid
{
    for (int r=grid.getNumRows()/2; r<grid.getNumRows(); r++)
    {
        for (int c=0; c<grid.getNumCols(); c++)
        {
            grid.setColor(r, c, Color.RED);
        }
    }
}
```



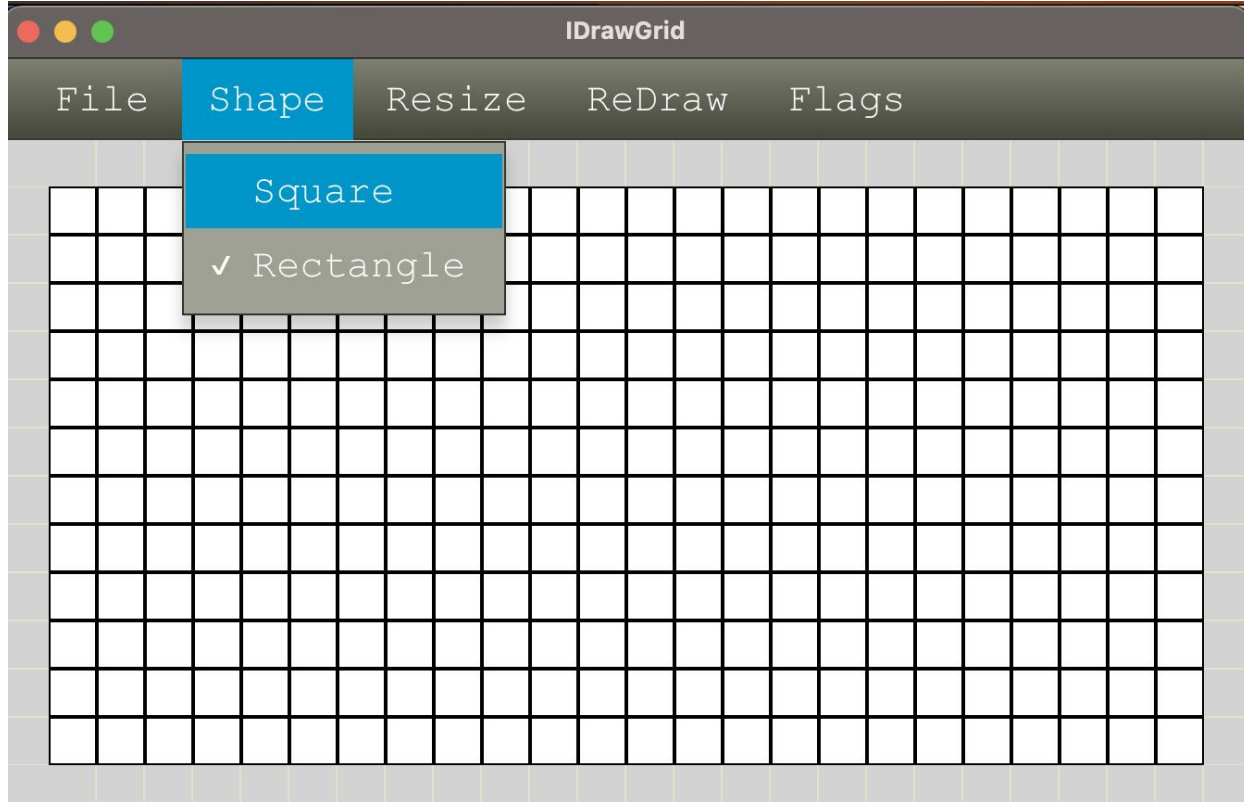
Diagonal1, second in the list

```
@Flag(name = "Diagonal1", num = 2)
public static void drawDiagonal1(IDrawGrid grid)
{
    for (int r=0; r<grid.getNumRows(); r++)
    {
        grid.setColor(r, r, Color.BLUE);
    }
}
```

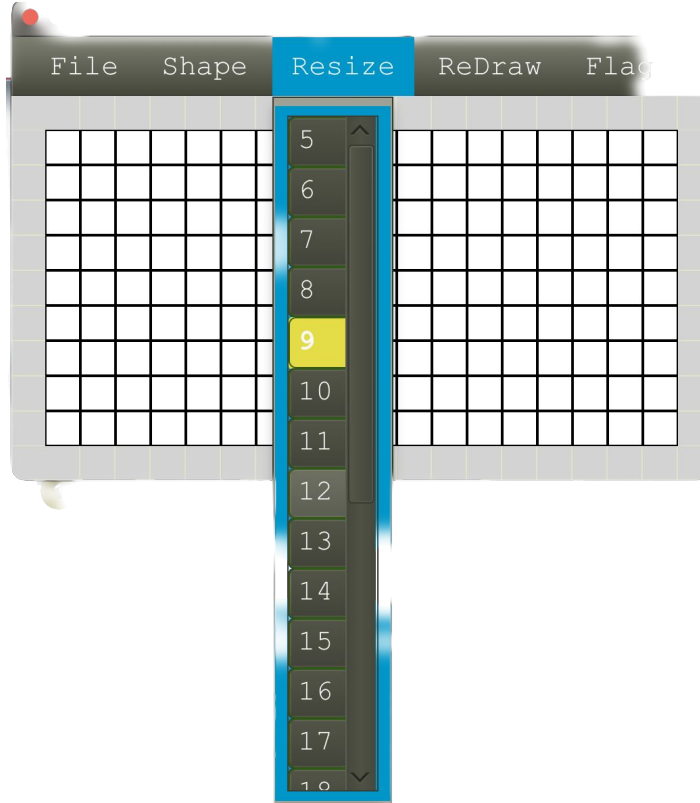


Square or Rectangular grid?

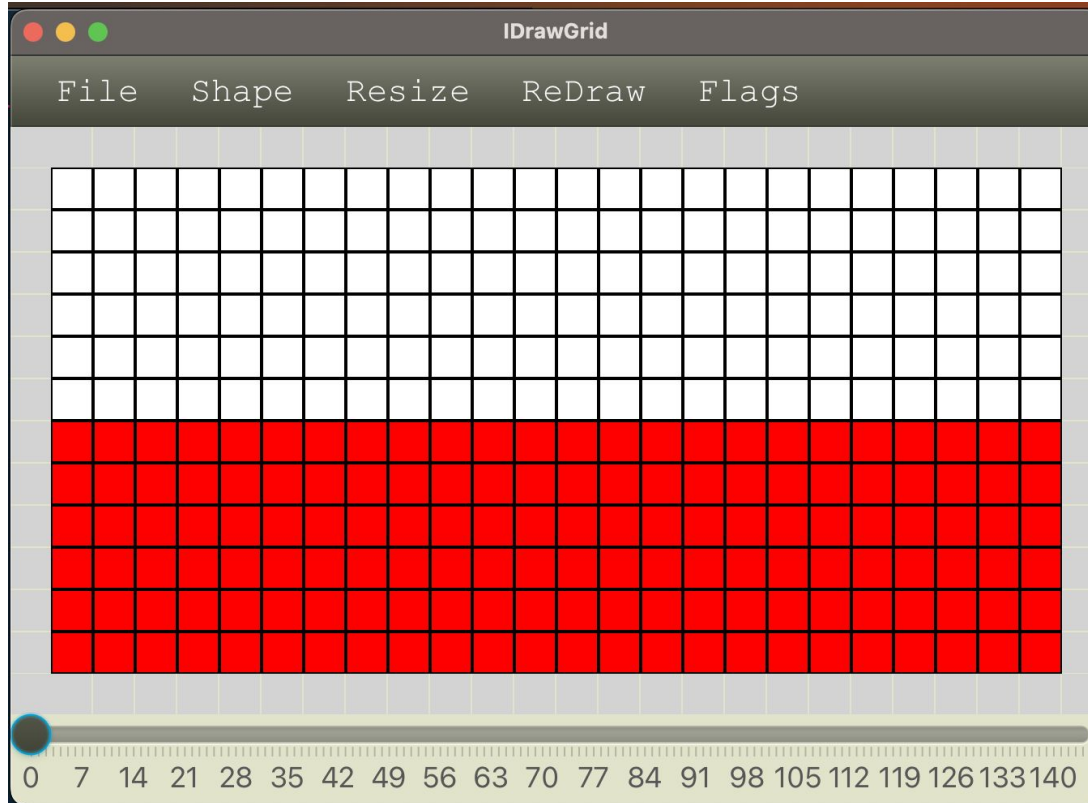
Rectangle is 2x as wide as tall



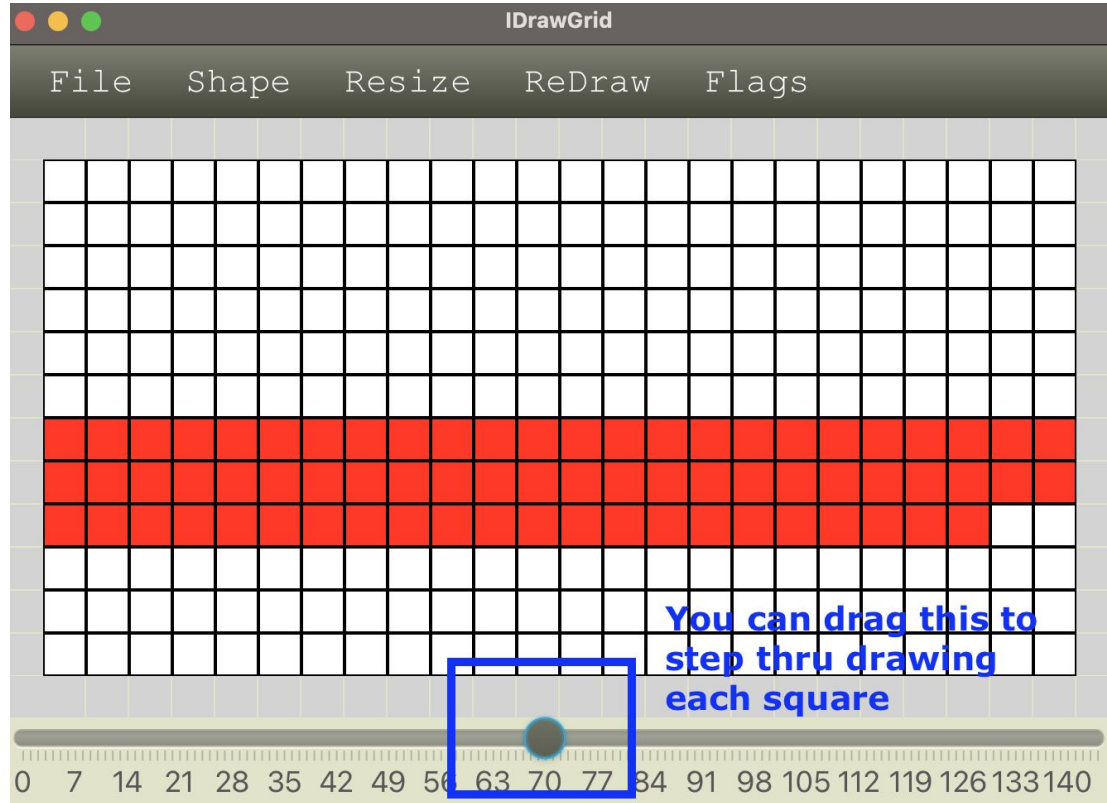
Change the size of the grid



We clicked "Poland" from the Flags menu!



We clicked Poland, and then we dragged the thumb at the bottom; this is the order we drew things



Check out the code for Poland again

This is a *function* and you are given an IDrawGrid

```
@Flag(name = "Poland", num = 1)
public static void drawPoland(IDrawGrid grid)
{
    for (int r=grid.getNumRows()/2; r<grid.getNumRows(); r++)
    {
        for (int c=0; c<grid.getNumCols(); c++)
        {
            grid.setColor(r, c, Color.RED);
        }
    }
}
```

What can we do with an IDrawGrid?

Only 3 methods

```
int getNumRows()
```

```
int getNumCols()
```

```
void setColor(int row, int col, Color color)
```

This project does a lot of things behind the scenes

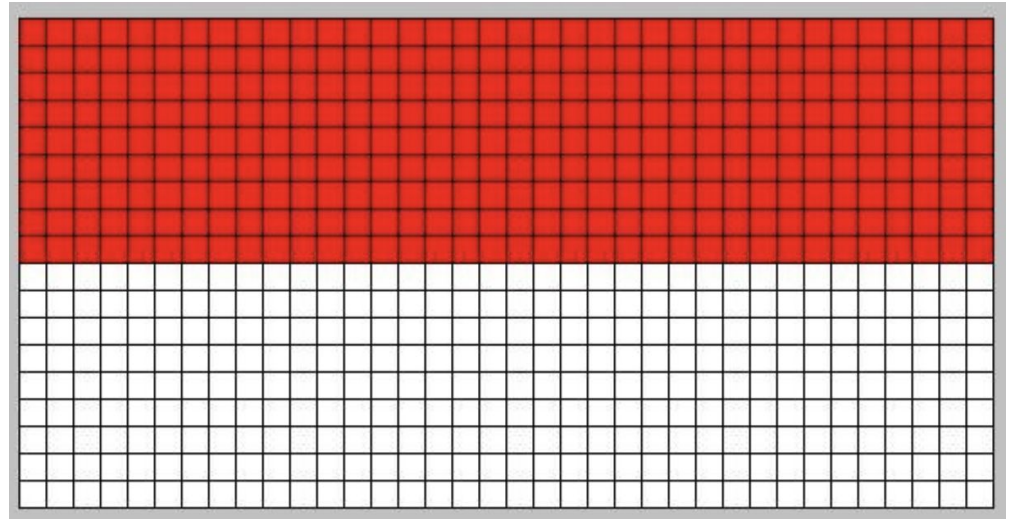
- Constructor calls
- Resizing
- Mapping functions to menu options
- Finding your code

This makes things easier for you to code, but can also be more confusing

For your lab this week, you will have blank functions to fill in

```
@Flag(name = "Indonesia", num = 2)
public static void drawIndonesia(IDrawGrid grid)
{
    // your code here
}
```

You fill in each method



Advice:

- RESTART THE PROGRAM after making changes to the code of a flag
 - It doesn't update automatically

Let's go through each of the other methods